

Options Pricing Using Heat Equation

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1 Introduction

This document presents the mathematical processes used to create a program that numerically computes option prices and calculates implied volatility.

The central result is that the Black-Scholes equation can be transformed into the heat equation through a change of variables, which allows the use of well known numerical methods on the heat equation to calculate European option prices.

2 Background

The holder of a European call option has the right, but not the obligation, to purchase a stock at the strike price K on a future expiration date T . A key distinction that applies only to European options is that they cannot be exercised before their expiration date. The question being investigated in this project is, "What is this option worth today?"

In 1973, this question was answered by Fischer Black, Myron Scholes, and Robert Merton by constructing a portfolio in which all randomness was eliminated by choosing a combination of the option and shares of the underlying stock and invoking the no arbitrage condition (a process known as delta hedging). This argument implied that the option price V must satisfy the following partial differential equation,

$$\frac{\partial V}{\partial t} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} + rS \frac{\partial V}{\partial S} - rV = 0 \quad (1)$$

where S is the current stock price, t is time, r is the risk free interest rate, and σ is the volatility of the stock.

Black and Scholes solved this equation by transforming it into the heat equation and applying the heat kernel to obtain the closed form solution,

$$C = SN(d_1) - Ke^{-rT}N(d_2) \quad (2)$$

where C is the price of a European call option today, $d_1 = \frac{\ln(S/K) + (r + \frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}$, $d_2 = d_1 - \sigma\sqrt{T}$, and N is the standard normal cumulative distribution function.

The goal of this project is to replicate this transformation but solve the heat equation numerically using the finite difference method rather than analytically with the heat kernel. Using numerical methods helps generalize to settings where no closed form solution exists. Additionally, the implied volatility is computed by inverting the Black-Scholes formula using Newton's method. Results are verified against the closed form solution to confirm accuracy.

3 The Transformation

Although it is possible to discretize the Black-Scholes equation directly, the variable coefficients S^2 and S require different update rules at each point on the grid. To eliminate these coefficients and simplify the numerical solving process, we transform the equation into the heat equation,

$$\frac{\partial u}{\partial \tau} = \alpha \frac{\partial^2 u}{\partial x^2} \quad (3)$$

where u , originally temperature, is the option value, x , originally position, is the price of the stock, α is thermal diffusivity, and τ , originally time, is a rescaled time variable running backwards from expiration to the present. This transformation is possible due to three key variable substitutions.

3.1 Substitution 1: $x = \ln(S/K)$

The first substitution replaces the current stock price S with the natural logarithm of the ratio of the current stock price and the strike price, $x = \ln(S/K)$.

We can then apply the multivariable chain rule to rewrite the derivatives in terms of x .

$$\begin{aligned} \frac{\partial V}{\partial S} &= \frac{1}{S} \frac{\partial V}{\partial x} \\ \frac{\partial^2 V}{\partial S^2} &= \frac{1}{S^2} \left(\frac{\partial^2 V}{\partial x^2} - \frac{\partial V}{\partial x} \right) \end{aligned}$$

After substituting into the Black-Scholes equation and simplifying, the coefficients S^2 and S cancel, leaving only constant coefficients and giving us the following equation.

$$\frac{\partial V}{\partial t} + \frac{1}{2}\sigma^2 \frac{\partial^2 V}{\partial x^2} + \left(r - \frac{1}{2}\sigma^2\right) \frac{\partial V}{\partial x} - rV = 0 \quad (4)$$

3.2 Substitution 2: $\tau = \frac{1}{2}\sigma^2(T - t)$

The second substitution replaces time t with a rescaled time variable $\tau = \frac{1}{2}\sigma^2(T - t)$. The purpose of this substitution is to reverse the direction of time, since the heat equation runs forward in time while the Black-Scholes equation runs backward in time.

Again, we apply the chain rule to get

$$\frac{\partial V}{\partial t} = -\frac{1}{2}\sigma^2 \frac{\partial V}{\partial \tau}$$

After substituting into Equation (4), moving the time derivative to the right hand side, dividing both sides by $\frac{1}{2}\sigma^2$, and letting $k = \frac{2r}{\sigma^2}$, the equation becomes

$$\frac{\partial V}{\partial \tau} = \frac{\partial^2 V}{\partial x^2} + (k - 1) \frac{\partial V}{\partial x} - kV \quad (5)$$

We still need to deal with the first derivative term and the zeroth order term, which the final substitution will aid us with.

3.3 Substitution 3: $V = (e^{\alpha x + \beta \tau})u$

To start, we rewrite the option price V as $(e^{\alpha x + \beta \tau})u$, where u is a new unknown function and α and β are constants that we will choose to eliminate the two terms left over from the previous step.

Using the product rule, we can find that

$$\begin{aligned} \frac{\partial V}{\partial \tau} &= e^{\alpha x + \beta \tau} \left(\beta u + \frac{\partial u}{\partial \tau} \right) \\ \frac{\partial V}{\partial x} &= e^{\alpha x + \beta \tau} \left(\alpha u + \frac{\partial u}{\partial x} \right) \\ \frac{\partial^2 V}{\partial x^2} &= e^{\alpha x + \beta \tau} \left(\alpha^2 u + 2\alpha \frac{\partial u}{\partial x} + \frac{\partial^2 u}{\partial x^2} \right) \end{aligned}$$

We can now clearly see that all the terms in the equation contain $e^{\alpha x + \beta \tau}$, allowing us to remove the exponential term entirely, resulting in the following equation.

$$\beta u + \frac{\partial u}{\partial \tau} = \alpha^2 u + 2\alpha \frac{\partial u}{\partial x} + \frac{\partial^2 u}{\partial x^2} + (k-1)\alpha u + (k-1)\frac{\partial u}{\partial x} - ku \quad (6)$$

It is now time to carefully choose the constants α and β such that the terms $\frac{\partial u}{\partial x}$ and u vanish, leaving us with the desired result.

We can start by collecting the $\frac{\partial u}{\partial x}$ terms on one side and factoring out $\frac{\partial u}{\partial x}$. Then, we solve for α so that the coefficient term equals 0

$$(2\alpha + k - 1)\frac{\partial u}{\partial x} = 0$$

$$2\alpha + k - 1 = 0$$

$$\alpha = \frac{1 - k}{2}$$

To find β , we start with a similar approach. We collect the u terms on both sides, factor out u , and substitute the α value found in the previous step.

$$\begin{aligned} \beta &= \alpha^2 + (k-1)\alpha - k \\ &= \left(\frac{1-k}{2}\right)^2 + (k-1)\left(\frac{1-k}{2}\right) - k \\ &= -\frac{(1+k)^2}{4} \end{aligned}$$

Using these values for α and β , we can eliminate the $\frac{\partial u}{\partial x}$ and u terms, leaving us with the heat equation.

$$\frac{\partial u}{\partial \tau} = \frac{\partial^2 u}{\partial x^2} \quad (7)$$

4 Initial Condition

When $\tau = 0$, the payoff value of a European call option is given by $\max(S - K, 0)$ ($\max(K - S, 0)$ for a put option). When we express a payoff in transformed variables, it serves as the initial condition $u_0(x)$ for the heat equation. Substituting $S = Ke^x$ into the call payoff and applying the inverse of the third change of variables at $\tau = 0$, we get

$$u_0(x) = \max\left(e^{\frac{1}{2}(k+1)x} - e^{\frac{1}{2}(k-1)x}, 0\right) \quad (8)$$

Similarly, the put payoff transforms to

$$u_0(x) = \max \left(e^{\frac{1}{2}(k-1)x} - e^{\frac{1}{2}(k+1)x}, 0 \right) \quad (9)$$

5 Finite Difference Method

To solve the heat equation numerically, we use the explicit finite difference method. The continuous domain is divided into N equally-spaced points with spacing Δx , and time is broken into steps of size $\Delta \tau$. Since exact derivatives cannot be computed on a discrete grid, we approximate the second spatial derivative to be

$$\frac{\partial^2 u}{\partial x^2} \approx \frac{u_{i+1} - 2u_i + u_{i-1}}{\Delta x^2} \quad (10)$$

At each time step n , for every interior grid point i , the new value u is

$$u_i^{n+1} = u_i^n + \frac{\Delta \tau}{\Delta x^2} (u_{i+1}^n - 2u_i^n + u_{i-1}^n) \quad (11)$$

For numerical stability, the time step must satisfy the following condition.

$$\frac{\Delta \tau}{\Delta x^2} \leq \frac{1}{2} \quad (12)$$

To see why, we rewrite the update rule as

$$u_i^{n+1} = (1 - 2r)u_i^n + ru_{i+1}^n + ru_{i-1}^n \quad (13)$$

where $r = \frac{\Delta \tau}{\Delta x^2}$. The new value is a weighted sum of the current point and its two neighbors. When $r > \frac{1}{2}$, the weight $1 - 2r$ becomes negative, which produces values that grow without bound and break the simulation.

6 Implied Volatility

In practice, the volatility σ of a stock is not observable. However, given a market price for an option, it is possible to work backwards to find the implied volatility.

If we are given a market price for the call option C_{market} , we want to find σ such that

$$C(\sigma) - C_{market} = 0 \quad (14)$$

To find σ , we can use Newton's method. We start from an initial guess for what we think σ could be, denoted as σ_0 . Newton's method updates σ at each step according to

$$\sigma_{n+1} = \sigma_n - \frac{C(\sigma_n) - C_{market}}{\nu} \quad (15)$$

where $\nu = \frac{\partial C}{\partial \sigma} = SN'(d_1)\sqrt{T}$ is the vega of the option and N' is the standard normal probability density function.

7 Results

The pricer was tested using the following parameters:

- Stock price $S = \$105$
- Strike price $K = \$110$
- Time to expiry $T = 0.5$ years
- Risk free interest rate $r = 5\%$
- Volatility σ drawn randomly from $U(0.1, 0.5)$

7.1 Option Prices

The numerical call price of \$10.18 differed from the analytical price of \$10.12 by 0.5364%. The numerical put price of \$12.37 differed from the analytical price of \$12.41 by 0.2615%. The put-call parity error was \$0.09.

Both prices are accurate to within 1% of the analytical solution, confirming that the heat equation approach correctly prices European options.

It also was observed that increasing the number of grid points reduced the error further, which is expected when using finite difference methods. This was not implemented in the final program in the interest of saving computing power.

7.2 Implied Volatility

Using the parameters above and an unknown randomly generated σ , Newton's method recovered an implied volatility of 37.79%. The program then revealed the true value of σ , which was 37.61%, producing an absolute error of 0.001835 (0.49%). Convergence was achieved in 3 iterations.